

Response to Reviewers' Comments for Manuscript [25000741]

Reviewer A

[Questions]

(1) Have you considered incorporating the transport time of AGVs into the model? Specifically, it might be beneficial to include a constraint on $K_{i,j}$ and $H_{i,j}$ with a parameter such as $t_{i,j,k}$, similar to equations (7) and (8).

Answer:

We sincerely appreciate the reviewer's insightful suggestion. In the current model, the AGV working time is calculated based on the difference between the finish time $H_{i,j}$ and the start time $K_{i,j}$ of handling each process (as shown in Equation (6)). While the current model implicitly includes transport time, we fully agree with your suggestion to explicitly model it for precision.

To implement this, we:

1. Added parameter $t_{i,j,k}^{trans}$ to represent AGV handling time of the process j of workpiece i to machine k .
2. Added constraints (11)-(12) to explicitly include $t_{i,j,k}^{trans}$.

Specific modifications in the manuscript:

From revised paper (Table 2):

$t_{i,j,k}^{trans}$ — Handling time of the process j of workpiece i to machine k

From revised paper (Section 2.2):

$$H_{i,j} = K_{i,j} + t_{i,j,k}^{trans} \quad \forall i = 1, 2, \dots, n \quad \forall j = 1, 2, \dots, s_i \quad \forall k = 1, 2, \dots, m \quad (11)$$

$$K_{i,j} \geq F_{i,j-1,k} + t_{i,j,k}^{trans} \quad \forall i = 1, 2, \dots, n \quad \forall j = 1, 2, \dots, s_i \quad \forall k = 1, 2, \dots, m \quad (12)$$

From revised paper (Section 2.2):

Constraint (11) and (12) defines the finish time of AGV handling as the sum of the start time and the handling time, and ensures that the start time of AGV handling for a process must account for the handling time required to move from the previous process's machine to the current process's machine;

(2) Could you clarify the necessity of describing the model using both “workpieces” and “processes”? Would it be possible to represent the model solely in terms of “processes”?

Answer:

Distinction and Necessity of "Workpieces" and "Processes":

Workpieces: In the paper, workpieces represent the parts or products to be processed. Each workpiece consists of multiple processes, and the introduction of workpieces helps clarify the overall processing flow of each part.

Processes: Processes refer to the specific processing steps of a workpiece on a machine. Each workpiece may contain multiple processes, and these processes are subject to precedence constraints. Hierarchical Relationship Between "Workpieces" and "Processes":

"Workpieces" and "processes" represent two hierarchical levels in the model. Workpieces are

at a higher level, encompassing all the processes required to complete a part, while processes are at a lower level, describing the detailed steps of machining and assembly.

This hierarchical structure allows the model to clearly describe both the overall processing flow of each workpiece and the detailed processing steps of each process.

If the model were represented solely in terms of workpieces, it would lack the granularity needed to describe the detailed processing steps (i.e., processes) of each workpiece, which are essential for scheduling and optimization.

Conversely, if the model were represented solely in terms of processes, it would be difficult to track the completion status of each workpiece, as the completion of a workpiece depends on the completion of all its associated processes.

The combination of "workpieces" and "processes" ensures that the model can accurately describe both the overall processing flow and the detailed processing steps, making it more comprehensive and easier to understand.

[Comments]

(1) In equation (7), it might be necessary to explicitly add the index constraints $i \in p$ and $j \in q$ to ensure correctness.

Answer:

We thank the reviewer for catching this important technical point. Since we have added an equation, Equation (7) is now changed to Equation (8). The constraints $i \neq p$ and $j \neq q$ have now been explicitly added to Equation (8) to prevent trivial solutions where a process would conflict with itself.

Specific modifications in the manuscript:

Revised Equation (8) in Section 2.2:

$$\begin{aligned} S_{p,q,k} &\geq S_{i,j,k} + t_{i,j,k} - L \cdot (1 - Y_{i,j,p,q}^k) \\ \forall i, p &= 1, 2, \dots, n \quad \forall j, q = 1, 2, \dots, s_i \\ \forall k &= 1, 2, \dots, m \quad i \neq p, j \neq q \end{aligned} \quad (8)$$

(2) In equation (9), the index constraint $\forall v = 1, 2, \dots, r$ may be unnecessary.

Answer:

We thank the reviewer for this observation that removing the constraint $\forall v = 1, 2, \dots, r$ from Equation (9) is reasonable. We will incorporate this modification in the revised manuscript. Since we have added an equation, Equation (9) is now changed to Equation (10).

Specific modifications in the manuscript:

Revised Equation (10) in Section 2.2:

$$\begin{aligned} \sum_{v=1}^r V_{i,j,v} &= 1 \\ \forall i &= 1, 2, \dots, n \quad \forall j = 1, 2, \dots, s_i \end{aligned} \quad (10)$$

(3) Could you clarify the definition of $agen_{\{i\}}$ used in equations (13) and (14).

Answer:

We appreciate the reviewer's attention to detail. The reviewer may have misunderstood $\alpha \cdot gen_i$ as a single variable or symbol. We apologize for this ambiguity and have revised the formulas in the manuscript to more clearly express the relationship between the α and gen_i to avoid ambiguity and

potential misunderstandings.

To eliminate ambiguity, we have:

1. Defined α as a positive parameter (adaptation rate coefficient) and gen_i as the current generation number in Section 3.2.3.

2. Rewritten equations (13)-(14) to show the multiplicative relationship more clearly. Since we have added some equations, Equation (13)-(14) are now changed to Equation (16)-(17).

Specific modifications in the manuscript:

Revised Equation (16)-(17) in Section 3.2.3:

$$P_{ac} = 0.8 + \alpha_{ci}e^{-\alpha \cdot gen} + \alpha \cdot \ln(1 + \delta_r \cdot rak_i) \quad (16)$$

$$P_{am} = 0.03 + \alpha_{mi}e^{-\alpha \cdot gen} + \alpha \cdot \ln(1 + \delta_r \cdot rak_i) \quad (17)$$

In the formula, the parameters α_{ci} and α_{mi} represent the crossover probability control coefficient and the mutation probability control coefficient, respectively. α and δ_r are positive parameters, and gen indicates the current iteration number of the population. rak_i represents the non-dominant rank of the crossover or mutation objects under the current iteration number. **The non-dominated ranks are sorted from the best solutions, where $rak_i=1$ represents the highest rank (first Pareto front), $rak_i=2$ denotes the second-best rank (second front), and so on.** Specifically, during crossover, rak_i represents the smaller non-dominant rank among the two selected individuals. If they are equal, one is randomly chosen. During mutation, rak_i represents the non-dominant rank of the selected individual itself.

(4) I recommend replacing the term “optimal” with “best” to avoid potential ambiguity. The term “optimal solution” typically refers to a mathematically proven optimal solution, whereas in this context, it seems to mean the best solution obtained by an algorithm.

>Fig. 17 shows a comparison of Gantt charts for the optimal makespan solutions obtained using IA-NSGA-II, NSGA-II, and IGA.

Answer:

We appreciate the reviewer's precise terminology suggestion. In our study, the so-called "optimal solution" indeed refers to the best solution obtained by the algorithm, rather than a mathematically proven global optimal solution. Therefore, using "best" is more accurate and avoids potential ambiguity. Since we have added a figure, Fig. 17 is now changed to Fig. 18.

Specific modifications in the manuscript:

From revised paper (Section 4.3):

1. **Fig. 18 shows a comparison of Gantt charts for the solutions obtained using IA-NSGA-II, NSGA-II, and IGA.**

2. From the figure, the result obtained using IA-NSGA-II is better than those obtained using NSGA-II and IGA. This is because NSGA-II tends to get trapped in local optima, preventing it from finding the **best** solution, while IGA is not effectively designed to handle dual-objective problems, leading to multiple changes in processing machines for processes.

(5) Figure 17 is somewhat difficult to read. It would be helpful to replace it with a vector image, ensuring that the numbers are clearly arranged for better readability.

Answer:

We appreciate the reviewer's feedback on visualization clarity. Since we have added a figure,

Fig. 17 is now changed to Fig. 18. The current figure may lack clarity in terms of numbers and layout, which could affect the reader's experience.

We thank the reviewer for this suggestion and have redesigned the layout of Figure 18 to ensure clearer visualization. Additionally, we have made corresponding adjustments to Figure 20 to achieve better clarity as well.

Specific modifications in the manuscript:

From revised paper (Section 4.3):

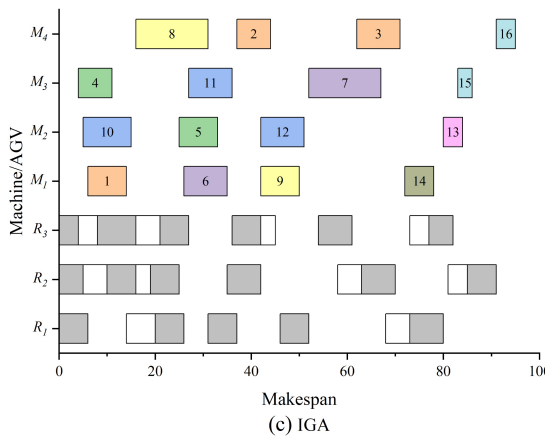
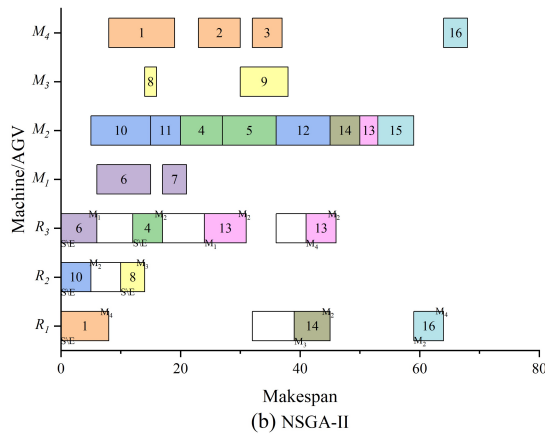
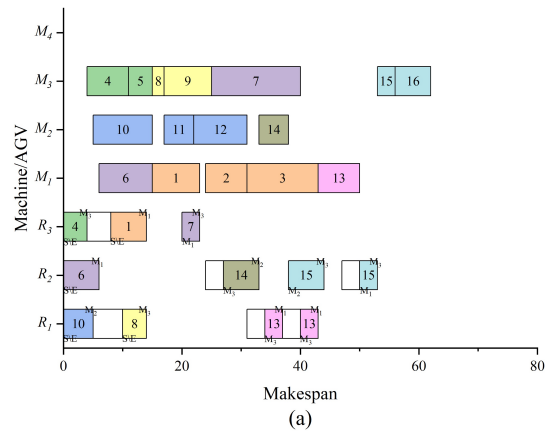


Fig. 18. Comparison of Gantt charts using IA-NSGA-II, NSGA-II, and IGA

From revised paper (Section 4.5):

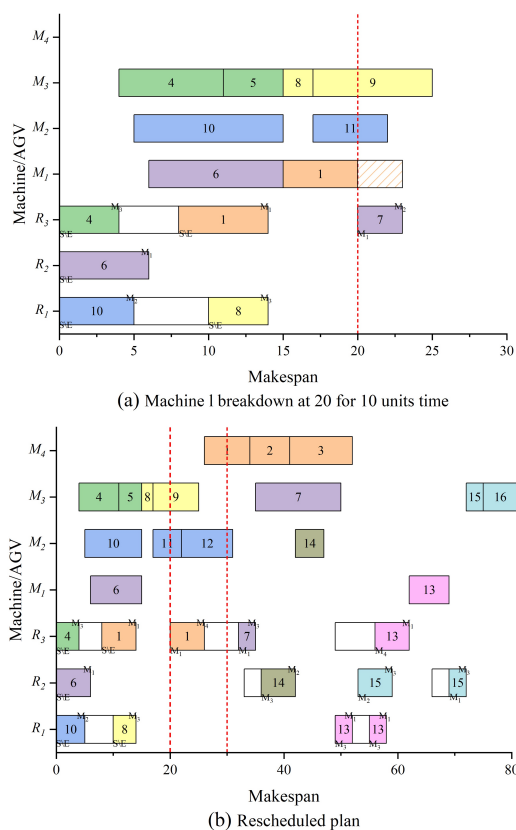


Fig. 20. Gantt charts before and after rescheduling

(6) The discussion on rescheduling may not be directly related to the optimization algorithm, which is the main focus of this paper. If this section is retained, I recommend providing numerical experiments to demonstrate the advantages of the proposed algorithm over other rescheduling methods. Otherwise, considering its removal might help maintain the focus of the paper.

Answer:

We understand the reviewer's concern. While the rescheduling section is not the core focus of this paper, it serves as a preliminary study for our future research on dynamic scheduling problems. We plan to explore dynamic scheduling in more depth in subsequent work, particularly focusing on rescheduling strategies for dynamic events such as machine breakdowns and rush orders. The rescheduling experiments in this paper are intended to provide initial validation of the proposed algorithm's potential in dynamic environments and lay the groundwork for future research.

The primary contribution of this paper is the proposed improved adaptive NSGA-II algorithm (IA-NSGA-II) for solving the FAJSP-AGVs problem. The rescheduling section demonstrates the algorithm's preliminary performance in dynamic environments.

Although this paper does not include detailed comparisons with other rescheduling methods, the rescheduling experiments verify the feasibility and stability of the proposed algorithm in dynamic environments.

We thank the reviewer for this suggestion and hope to retain the rescheduling section. In the revised manuscript, we will further clarify the positioning of this section and emphasize its role in laying the groundwork for subsequent research.

Specific modifications in the manuscript:

From revised paper (Section 3.5):

In this section, we present a preliminary study on rescheduling strategies, which serves as a foundation for our future research on dynamic scheduling problems. While the primary focus of this paper is on the optimization algorithm (IA-NSGA-II) for the FAJSP-AGVs, the rescheduling experiment demonstrate the potential of the proposed algorithm in dynamic environment. For the future work, we will further explore rescheduling strategies under various dynamic events, such as machine breakdowns and rush orders.

From revised paper (Section 4.5):

In this section, we evaluate the performance of the proposed IA-NSGA-II through a series of experiments. Additionally, we include a preliminary rescheduling experiment to demonstrate the algorithm's potential in dynamic environments. This experiment lays the groundwork for our future research on dynamic scheduling problems.

From revised paper (Conclusions):

Rescheduling experiment preliminarily validate the method's stability in handling dynamic event of machine breakdowns.

For the future work, we could explore diverse application scenarios, such as distributed assembly shop scheduling, AGV conflict-free path planning, and other dynamic scheduling contexts, as well as the integration with advanced optimization techniques such as decomposition-based methods (MOEA/D), differential evolution (DE), and their hybrid approaches (e.g., MOEA/D-DE) [28].

Reviewer B

[Major Comments]

major comment 1 (Title)

Table 1 shows both of the optimization problem and algorithm of this paper differs from the existing papers. Based on Table 1, the originality is in the contribution of this paper is both of the optimization problem and algorithm, but I feel the title is algorithm's proposal. Please correct the title to an appropriate name: for instance, "Multi-objective comprehensive scheduling problem of flexible assembly job shop problem with AGVs and NSGA-II based optimization algorithm".

Answer:

We fully agree with the reviewer's suggestion. The contributions of this paper lie not only in proposing an Approach (IA-NSGA-II) but also in establishing a new optimization problem model, namely the Flexible Assembly Job Shop Scheduling Problem with AGVs (FAJSP-AGVs). Therefore, the title should reflect both the originality of the optimization problem and the method. After careful consideration, we will follow the reviewer's advice to refocus the title on the problem description and solution approach (rather than emphasizing the algorithm name).

We thank the reviewer for this suggestion and will revise the title in the manuscript to more comprehensively reflect the contributions of the paper.

Specific modifications in the manuscript:

From revised paper (Title):

Multi-objective comprehensive Approach for flexible assembly job shop scheduling problem with AGVs

major comment 2 (Page3, Equations (1), (2), and (5))

This paper proposes the three-objective optimization problem as the FAJSP-AGVs but Equations (1), (2), and (5) show minimization problems independent of each other. Please rewrite these as follows: "minimize (Cmax, Et, Tv)...(1)", "a. Makespan: Cmax = max(...)...(2)", "b. Total machine energy consumption: Et = Epro + Est...(3)", and "c. Total AGV working time: Tv= ...(6)".

Answer:

We thank the reviewer for catching this inconsistency in our mathematical presentation. The current equations do not sufficiently reflect the characteristics of multi-objective optimization, and the revised equations will more clearly express the three-objective optimization problem proposed in this paper.

We thank the reviewer for this suggestion and will modify these Equations in the revised manuscript to more clearly express the three-objective optimization problem.

Specific modifications in the manuscript:

Since we have added some equations, Equation (1)(2)(5) are now changed to Equation (2)(3)(6).

Revised Equation (1)(2)(3)(6) in Section 2.2:

$$\text{minimize } (C_{max}, E_t, T_v) \quad (1)$$

$$C_{max} = \max(C_i \mid i = 1, 2, \dots, n) \quad (2)$$

$$E_t = E_{pro} + E_{st} \quad (3)$$

$$T_v = \sum_{i=1}^n \sum_{j=1}^{s_i} (H_{ij} - K_{ij}) \quad (6)$$

From revised paper (Section 2.2):

Where, formula (1) represents the multi-objective optimization problem, which aims to minimize the makespan (C_{max}), total machine energy consumption (E_t), and total AGV working time (T_v); formula (2) represents the makespan; Formula (3) represents the total energy consumption; Formula (4) represents the machining energy consumption; Formula (5) represents the machine idle energy consumption; Formula (6) represents the AGV working time, which is composed of two parts: unloaded movement and loaded handling;

major comment 3 (Table 2)

Table 2 shows the symbolic description of model parameters and the FAJSP-AGVs should be formulated to the mixed-integer linear programming problem (MILP). But I guess the reader cannot distinguish between the decision variables and fixed parameters in the FAJSP-AGVs. Therefore, please add a description of decision variables under Equation (12) as follows: "The FAJSP-AGVs is formulated to the mixed-integer linear programming problem (MILP). The integer variables $X_{\{i,j,k\}}$, $Y^k_{\{i,j,p,q\}}$, $V_{\{i,j,v\}}$, and $Z_{\{i,j,p,q,v\}} \in \{0, 1\}$; and the continuous variables $S_{\{i,j\}}$, $S_{\{p,q,k\}}$, $S_{\{i,j,k\}} \in \mathbb{R}$ are mixed. Number of the integer variables is $\sim$ and number of the continuous variables is $\sim$."

Answer:

We fully agree with the reviewer's suggestion. To help readers better understand the model, we will add a description under Equation (15) to clarify that the FAJSP-AGVs is formulated as a mixed-

integer linear programming problem (MILP) and list the types and numbers of decision variables.

Specific modifications in the manuscript:

From revised paper (Section 2.2):

The FAJSP-AGVs is formulated as a mixed-integer linear programming problem (MILP). The integer variables $X_{ij,k}$, $Y_{ij,p,q}^k$, $V_{ij,v}$, and $Z_{ij,p,q,v} \in \{0,1\}$; and the continuous variables S_{ij} , $S_{p,q,k}$, $S_{ij,k} \in \mathbb{R}$ are mixed. The number of integer variables is $n \times s_i \times m + n \times s_i \times n \times s_i \times m + n \times s_i \times r + n \times s_i \times n \times s_i \times r$, and the number of continuous variables is $n \times s_i + n \times s_i \times m + n \times s_i \times m$.

major comment 4 (Chapter 3)

One of the contributions of this paper is the proposal of an algorithm. While the ideas are described from Subsections 3.1 to 3.3, the reader cannot understand an overall proposed NSGA-II. Please show flowchart or steps of the proposed NSGA-II in Chapter 3.

Answer:

We thank the reviewer for highlighting this critical need for algorithmic clarity. To help readers better understand the overall flow of the proposed improved adaptive NSGA-II algorithm (IA-NSGA-II), we will add a flowchart in Chapter 3. This will help readers better understand the proposed improved adaptive NSGA-II algorithm (IA-NSGA-II).

Specific modifications in the manuscript:

From revised paper (Section 3.4):

3.4 Algorithm Flowchart The improved adaptive NSGA-II algorithm (IA-NSGA-II) proposed in this paper integrates an adaptive genetic operation mechanism and VNS method to enhance the performance of the original NSGA-II. The algorithm begins with the initialization of a population using a process constraint matrix-based encoding method, followed by fitness evaluation, non-dominated sorting, and crowding distance calculation. Genetic operations, including improved adaptive crossover and mutation, are then applied to generate offspring populations.

To further expand the solution space, a VNS method is incorporated, which systematically explores different neighborhood structures. The parent and offspring populations are merged, and environmental selection is performed to preserve non-dominated solutions and maintain population diversity. This process iterates until the termination condition is met.

The overall flowchart of the proposed algorithm is shown in Fig. 11.

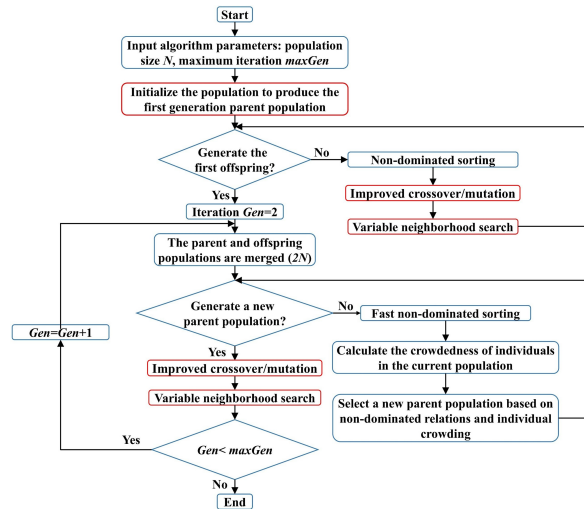


Fig. 11. Flowchart of the IA-NSGA-II

major comment 5 (Chapter 4)

Chapter 4 shows the usability of the proposed NSGA-II (IA-NSGA-II) by comparing it with the results of INSGA-II and IGA, but I guess the reader cannot understand the usability created by this feature. While this paper says "In this paper, the genetic operator is improved with the same purpose and further improved for the problem of local optimum.", it does not prove the effect of avoid the population falling into local optimum by comparing with/without elements. Without this, it is difficult to understand how it is a meaningful idea. Therefore, please add additional comparison results shown below:

- Crossover and mutation probability are fixed to a certain value in the proposed NSGA-II to verify the effect of adaptive crossover and mutation probability in 3.2.3.

- The Variable neighborhood search (VNS) is excluded from the proposed NSGA-II to verify the effect of VNS in 3.3.

Answer:

We sincerely appreciate the reviewer's suggestion and will incorporate an ablation study in the revised manuscript to validate the effects of both the adaptive crossover/mutation probability and VNS. This addition will help readers better understand the significance of the proposed algorithmic improvements.

Specific modifications in the manuscript:

From revised paper (Abstract):

An ablation study further demonstrates the contributions of the adaptive probability mechanism and VNS, highlighting their roles in enhancing solution quality and avoiding local optima.

From revised paper (Chapter 1):

(5) Several experiments, including an ablation study, validate the effectiveness of the proposed algorithm in improving solution quality and avoiding local optima.

From revised paper (Section 4.4):

4.4 Experiment 3 To further validate the effectiveness of the proposed improvements in the IA-NSGA-II, we conducted an ablation study to evaluate the contributions of the adaptive crossover/mutation probability and the VNS. The experiments were conducted using the same test instances and parameter settings as in Experiment 2 to ensure reliability and consistency.

The ablation study compares the following three variants of the algorithm:

- IA-NSGA-II: The complete algorithm with adaptive crossover/mutation probability and VNS.
- IA-NSGA-II (Fix): The algorithm with VNS but fixed crossover and mutation probabilities ($P_{ac}=0.8$, $P_{am}=0.03$).
- IA-NSGA-II(Non-VNS): The algorithm with adaptive crossover/mutation probability but without VNS.

The results of the ablation study are shown in Fig. 19. Comparing IA-NSGA-II and IA-NSGA-II (Fix), we observed that the adaptive probability mechanism significantly improved the algorithm's performance. The adaptive probability adjusts dynamically based on the population's state, improving exploration and exploitation. In contrast, the fixed probability variant struggled to maintain diversity in later iterations, resulting in inferior solutions. For example, the final makespan

achieved by IA-NSGA-II was 55, while IA-NSGA-II (Fix) achieved a final makespan of 60.

Comparing IA-NSGA-II and IA-NSGA-II(Non-VNS), we found that VNS played a crucial role in avoiding local optima. The VNS mechanism expanded the search space by systematically changing neighborhood structures, enabling the algorithm to escape from local optima and find better solutions. For example, the final makespan achieved by IA-NSGA-II was 55, while IA-NSGA-II(Non-VNS) achieved a final makespan of 62.

The ablation study demonstrates that both the adaptive crossover/mutation probability and the variable neighborhood search are essential components of the proposed IA-NSGA-II. The adaptive probability mechanism enhances the algorithm's ability to dynamically adjust its search strategy, while VNS ensures that the algorithm can explore a wider search space and avoid local optima. These improvements collectively contribute to the good performance of IA-NSGA-II in solving the FAJSP-AGVs.

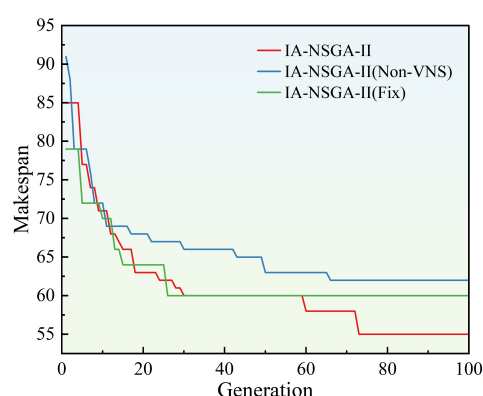


Fig. 19. Ablation study results for IA-NSGA-II variants

From revised paper (Conclusions):

Finally, feasibility and effectiveness comparison experiments were conducted using test cases proposed in the literature, including ablation study to validate the individual contributions of key algorithm components. The results demonstrate that the proposed method effectively solves the FAJSP-AGVs, with the ablation study confirming the critical roles of both the adaptive crossover/mutation probability and VNS.

[Minor Comments]

Although the author does not have to revise everything, please refer to the following minor comments for improving the current manuscript:

minor comment 1 (Section 4.1)

- "the original scheduling method proposed in the literature is referred to as "INSGA-II.", but the reference number of INSGA-II should be clearly written. Is it the literature [23]?
- "The scheduling method proposed in the literature is referred to as "IGA.", but the reference number of IGA should be clearly written. Is it the literature [27]?

Answer:

We appreciate the reviewer's attention to citation accuracy. To more clearly present the sources of the compared algorithms, we will explicitly indicate the corresponding reference numbers for

"INSGA-II"(literature [22]) and "IGA"(literature [27]) in the revised manuscript.

Specific modifications in the manuscript:

From revised paper (Chapter 1):

Wu et al. [22]	Makespan and energy consumption	Improved NSGA-II
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From revised paper (Section 4.1):

Through numerical experiments, we verify the usability of the improved adaptive NSGA-II scheduling method (IA-NSGA-II) proposed in this paper. Experiments 1 and 2 deal with static scheduling problem, while Experiment 3 deals with dynamic scheduling problem. In Experiment 1, we select a test benchmark instance from the reference Wu et al. [22] and compare the results of IA-NSGA-II with the original scheduling method “INSGA-II” proposed in Wu et al. [22]. In Experiment 2, we select a test benchmark instance from Wu et al. [27] in the context of FAJSP-AGV and compare the results of IA-NSGA-II with those of the scheduling method “IGA” proposed in Wu et al. [27] and traditional NSGA-II.

minor comment 2 (Section 4.1)

In Section 4.1, the correspondence between the outline of each experiment, the instances, the comparison method, and the parameter settings is a bit vague and misleading. In my opinion, after a brief description of the three experiments in section 4, each information should be listed in correspondence with the experiment number. Or it would be better to move them to the beginning of each section. For example, please refer to the following my draft: "Through numerical experiments, we verify the usability of the improved adaptive NSGA-II scheduling method (IA-NSGA-II) proposed in this paper. Experiments 1 and 2 deal with static scheduling problem, while Experiment 3 deals with dynamic scheduling problem. In Experiment 1, we select a test benchmark instance from the reference Wu et al. [22] and compare the results of IA-NSGA-II with the original scheduling method “INSGA-II” proposed in the literature [23]. In Experiment 2, we select a test benchmark instance from Wu et al. [27] in the context of FAJSP-AGV and compare the results of IA-NSGA-II with those of the scheduling method “IGA” and NSGA-II proposed in In Experiments 1 and 2, the solution was obtained independently for each instance to ensure reliability and consistency; Experiment 1 was run 10 times and Experiment 2 was run 20 times. In Experiment 3, three of the 20 scheduling results from Experiment 2 were randomly selected and the rescheduling method described in Section 3.4 was used. The common parameters for the algorithm in all experiments were set as follows: initial population size 100, number of genetic generations 100. The algorithm is implemented using the programming language MATLAB."

Answer:

We fully agree with the reviewer's suggestion. To more clearly present the design and parameter settings of each experiment, we will reorganize the content of Section 4.1 in the revised manuscript according to the reviewer's recommendation, ensuring that the instances, comparison methods, and parameter settings correspond clearly to each experiment number. This will help readers better understand the experimental design and the performance comparison of the algorithms.

Specific modifications in the manuscript:

From revised paper (Section 4.1):

Through numerical experiments, we verify the usability of the improved adaptive NSGA-II scheduling method (IA-NSGA-II) proposed in this paper. Experiments 1 and 2 deal with static

scheduling problem, while Experiment 3 deals with dynamic scheduling problem. In Experiment 1, we select a test benchmark instance from the reference Wu et al. [22] and compare the results of IA-NSGA-II with the original scheduling method “INSGA-II” proposed in Wu et al. [22]. In Experiment 2, we select a test benchmark instance from Wu et al. [27] in the context of FAJSP-AGV and compare the results of IA-NSGA-II with those of the scheduling method “IGA” proposed in Wu et al. [27] and traditional NSGA-II.

In Experiments 1 and 2, the solution was obtained independently for each instance to ensure reliability and consistency; Experiment 1 was run 10 times and Experiment 2 was run 20 times. In Experiment 3, three of the 20 scheduling results from Experiment 2 were randomly selected and the rescheduling method described in Section 3.5 was used. The common parameters for the algorithm in all experiments were set as follows: initial population size 100, number of genetic generations 100. The algorithm is implemented using the programming language MATLAB.

minor comment 3 (Figure 12)

Even though Figure 12 shows the results of a search with 100 population size, why are only 7 solutions obtained by IA-NSGA-II (Red Marker) and 6 solutions obtained by INSGA-II (Blue Marker) shown? Please add a description of Figure 12.

Answer:

We appreciate the reviewer's attention to result details. Since we have added a figure, Fig. 12 is now changed to Fig. 13. The reason for displaying only a subset of solutions in Figure 13 is as follows: In reference [22] (INSGA-II), the authors provided only 6 non-dominated solutions (Pareto front) in their results. To ensure a fair comparison with reference [22], we also display 6 solutions for INSGA-II (blue markers) in Figure 13. Our algorithm, IA-NSGA-II, generated more non-dominated solutions, but to directly compare with the results of INSGA-II, we selected the most representative 7 solutions (red markers) for display. Specifically, the leftmost red marker represents a solution where IA-NSGA-II achieves a better tardiness than the best tardiness reported in reference [22] for INSGA-II, along with the corresponding energy consumption at this tardiness. This result demonstrates that IA-NSGA-II not only outperforms in terms of energy consumption but also finds shorter tardiness compared to INSGA-II.

We thank the reviewer and will add a detailed description of Figure 13 in the revised manuscript, explaining the number of solutions and the selection criteria. This will help readers better understand the performance comparison of the algorithms.

Specific modifications in the manuscript:

From revised paper (Section 4.2):

Fig. 13 illustrates the comparison results of total machine energy consumption using IA-NSGA-II and INSGA-II for the same tardiness. **In Fig. 13, we present the non-dominated solutions (Pareto front) obtained by IA-NSGA-II (red markers) and INSGA-II (blue markers) with a population size of 100.** To ensure a fair comparison with the results from Wu et al. [22], we display 6 solutions for INSGA-II and 7 solutions for IA-NSGA-II. Notably, the leftmost red marker represents a solution where IA-NSGA-II achieves a better tardiness than the best tardiness reported in Wu et al. [22] for INSGA-II, along with the corresponding energy consumption at this tardiness. This result demonstrates that IA-NSGA-II not only outperforms in terms of energy consumption but also finds shorter tardiness compared to INSGA-II.

minor comment 4 (Chapter 4)

This paper improves some genetic operations in NSGA-II, but I guess the reason for the room for improvement in results may lie in GA or non-dominated sorting. There are multi-objective optimization methods: for instances, decomposition-based method (MOEA/D) or differential evolution (DE). Especially, MOEA/D-DE[28] combining MOEA/D and DE is known as the best multi-objective optimization method. In the future, the authors should compare a wide range of methods, including such state-of-the-art methods, to find an efficient way to solve the FAJSP-AGVs.

[Additional Literature]

[28] H.Li and Q. Zhang: "Multiobjective optimization problems with complicated Pareto sets, MOEA/D and NSGA-II", IEEE Transactions on Evolutionary Computation, Vol.13, No.2, pp.284-302 (2008)

Answer:

We sincerely appreciate the reviewer's insightful suggestion to broaden our comparative analysis. Although the improved adaptive NSGA-II algorithm (IA-NSGA-II) proposed in this paper performs well in solving the FAJSP-AGVs, we acknowledge that the genetic algorithm and non-dominated sorting may have certain limitations. The advanced multi-objective optimization methods mentioned by the reviewer, such as MOEA/D, DE, and MOEA/D-DE[28], indeed provide important directions for future research.

We thank the reviewer for the suggestion and will add a discussion on future research directions in the revised manuscript, explicitly mentioning comparative studies with other multi-objective optimization methods, such as MOEA/D, DE, and MOEA/D-DE. This will provide important directions for future research and help further optimize solutions for the FAJSP-AGVs.

Specific modifications in the manuscript:

From revised paper (Conclusions):

For the future work, we could explore diverse application scenarios, such as distributed assembly shop scheduling, AGV conflict-free path planning, and other dynamic scheduling contexts, as well as the integration with advanced optimization techniques such as decomposition-based methods (MOEA/D), differential evolution (DE), and their hybrid approaches (e.g., MOEA/D-DE) [28].

From revised paper (References):

(28) H.Li and Q. Zhang: "Multiobjective optimization problems with complicated Pareto sets, MOEA/D and NSGA-II", IEEE Transactions on Evolutionary Computation, Vol.13, No.2, pp.284-302 (2008)